

Ryan Wetzstein

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EDUCATION

North Dakota State University - Fargo, North Dakota Expected **Fall 2027**

Bachelor of Science in Computer Science | Minor: Artificial Intelligence **GPA: 3.44**

Dean's List: 4 semesters (Fall 2023 – Spring 2026)

TECHNICAL SKILLS

- Languages: Java, Python, SQL, TypeScript, C#
- Frameworks/Tools: React, Vite, PostgreSQL, Tauri, TensorFlow/Keras
- Concepts: Agentic Development, OOP, Agile/Scrum, Relational Databases, Neural Networks, Machine Learning

PROJECTS

Nexis - Open-source AI-native terminal & IDE May 2026 – Current

Rust, Tauri, React, TypeScript, xterm.js, CodeMirror 6 | nexisdev.org

- Directed an agentic-assisted rewrite inspired by an open-source terminal project into a full cross-platform IDE, authoring 285+ commits over 4 months across a Rust/Tauri backend and React/TypeScript frontend.
- Authored the React 19 / TypeScript frontend, xterm.js terminal, CodeMirror 6 editor, and file explorer, and designed a sandboxed, approval-gated AI agent system spanning 12+ LLM providers, with OS-keychain credential storage and SSRF/DNS-rebinding protections.
- Designed a full-stack systems architecture and created CI workflows for testing, containerizing, and releasing . Packaged application remains under 50MB with zero telemetry.

2026 NDSU Capstone - FarmQA January 2026 – May 2026

React, Vite, Microsoft Azure Blob Storage, Azure Functions, SQL Server | farmqa.com

- Worked in a team of 4 to deliver and present a client-approved full-stack application for FarmQA's internal team to visualize metrics and provide more information on sensitive data.
- Designed and implemented an SQL staging layer, populated by nightly scheduled jobs pulling from Azure Blob Storage, replacing runtime iteration over the full blob store with indexed queries, reducing load times ~98% (75s - ~1.5s) against ~250GB of test data.
- Did this as participation in the NDSU CSCI Capstone program; presented the finished product at the 2026 NDSU Engineering Senior Design Expo.

Music Genre CNN - Audio classification with CNNs Spring 2026 - *Artificial Intelligence Class Project*

Python, TensorFlow/Keras, librosa, NumPy | github.com/rwetz/music-genre-cnn

- Built an end-to-end pipeline converting raw audio into mel-spectrograms and trained/compared three CNN architectures (baseline, batch-norm, global-average-pooling), evaluating with confusion matrices and automatic checkpointing.

ArcGIS Parcel Harvester Fall 2026 - *Internship Project*

Python, ArcGIS REST API, pytest | <https://github.com/rwetz/arcgis-parcel-harvester>

- Generalized a South Dakota-specific scraper into a configurable, resumable harvester covering all 50 states with a rate-limited retry client and object-ID batch paging, classifying layers into 12 farmland categories across 5 government GIS sources; covered by 231 automated tests in CI.

WORK EXPERIENCE

Lemhi Technologies - Startup Company May 2026 – August 2026

Full-Stack Development Intern | Remote

- Built arc-gis-harvester, a resumable Python scraper for public ArcGIS REST servers, to construct a South Dakota farmland dataset underpinning the company's product; classified layers across agricultural, parcel, and boundary categories from 3+ state and federal GIS sources.
- Collaborated directly with startup founders on product scope and system selection, including creating user-flow diagrams for proposed solutions.